

Assignment 1

1. Explain the difference between frontend, backend, and full-stack development with

suitable real-world examples.

frontend=>UI design=> which is visible at client screen for example=>we visit any website which is visible to him on the website that is known as UI

Backend=>the code which provides functionality to the website and the user fetches data and saves data and this is not visible to the user; this is known as backend.

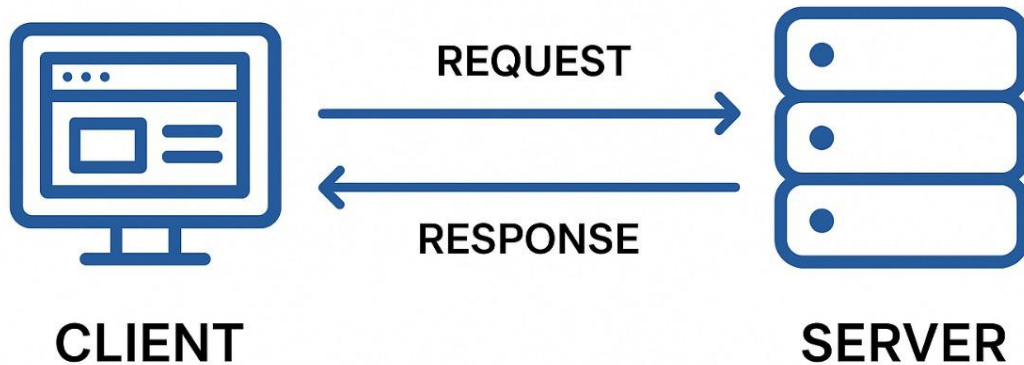
Full-stack=> frontend & backend combination is known as full-stack.

example=>

we go to any website which includes functionality and User Interface .

Question 2: Create a simple diagram showing how the client-server model works in web architecture.

CLIENT-SERVER MODEL



Question 3. Describe how a browser requests and displays a web page from a web server.

Browser find IP address of the server.

Browser only know html and css language if we write our code in any language than browser convert it into html .

Server sends requested html.

Browser processes html css pages.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Answer=

first we need an code editor (like vs code)

and we need browser to show preview of our code (like chrome aur firefox)

5. Explain what a web server is and give examples of commonly used servers.

answer=

web server is a kind of software use to send web pages to the browsers over the internet ,it helps in handling the requests ,processes them and send feedback to the requested content.

6. Define the roles of a frontend developer, backend developer, and database administrator

in a project.

Answer=>

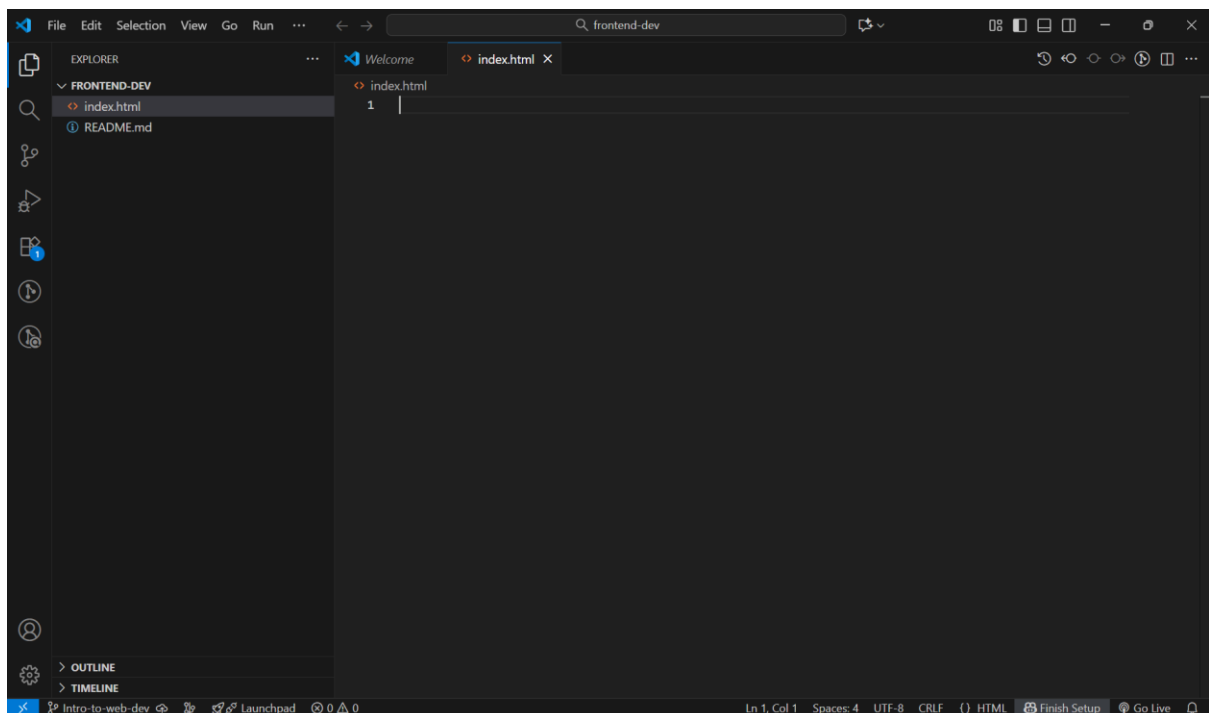
frontend Developer=> Design and implement the user interface using HTML,CSS,Javascript,React,Angular,Typescript etc of the website.

Backend Developer=>Build the working logic of website using Nodejs,Spring boot and maintain that.

DataBase administrator=>Manage and optimize the Database for efficient data storage.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a

screenshot of the setup.



8. Explain the difference between static and dynamic websites. Provide an example of each.

answer=

static => a website which is create for fix size screen and cannot be adjust according to my screen.

dynamic=>a website which is create for all size screens and adjust himself according to our screen is known as dynamic website.

9. Research and list five web browsers. Explain how rendering engines differ between them.

answer=>

- Google Chrome: Uses Blink, optimized for speed and performance.

- Mozilla Firefox: Uses Gecko, prioritizes web standards and privacy.
- Apple Safari: Uses WebKit, optimized for Apple hardware and energy efficiency.
- Microsoft Edge: Uses Blink (Chromium-based), combining Chrome's speed with Windows integration.
- Opera: Uses Blink, Chromium-based with unique features like a built-in VPN.

10. Draw a labeled diagram showing the basic web architecture flow — client, server,

database, and APIs.

Answer=>

GLA UNIVERSITY 	
Academic Dashboard Registration Main Account Policies My Advisor Academic Feedbacks Fee Details Value Added Course Feedback Panel GLAMS Notification Exam View Result CPI Calculator Online Exam Placement My Drive Drive Questions Placement Profile PREP Report	Attendance  100% BCSC 1012 (Theory) DAA Lekhraj 